



PAGES
PANGAEA GLOBAL EVENTS



ICMLAI-2026

International Conference on
Machine Learning & Artificial Intelligence

Advancing Intelligent Systems. Empowering the Future.

Title: Explainability-Aware Model Selection for MCI Subtype Classification in Brain MRI: Balancing Diagnostic Accuracy and Clinical Interpretability in Deep Learning

Authors: Ahana Priyanka N, Anushya V, Faatina S

AIM

To develop an optimized deep learning model with XAI-based explainability for multi-class classification of Alzheimer's Disease stages (EMCI, LMCI, MCI) using MRI images

INTRODUCTION

- Alzheimer's Disease is a progressive neurodegenerative disorder and a leading cause of dementia
- Early classification of EMCI, LMCI, and MCI is crucial for timely treatment
- MRI provides non-invasive brain imaging for detecting structural changes
- Deep learning models highlight relevant brain biomarker regions along with XAI it is efficient to understand the changes by clinicians
- However, these models require optimization of components to perform better classifications

LITERATURE SURVEY

Related Works:

- [El-Assy et al., 2024](#) - Proposed a CNN-based architecture for multi-class Alzheimer's classification using MRI, achieving high accuracy but focusing mainly on architecture design
- [Ali et al., 2024](#) - Developed a multistage CNN framework for Alzheimer's detection and staging, improving classification performance but lacking component-level optimization
- [Zarei et al., 2024](#) - Combined CNN with radiomic features for MRI-based classification of AD and MCI, showing strong accuracy but limited explainability
- [Altwijri et al., 2023](#) - Introduced a deep learning model for automatic Alzheimer's diagnosis from MRI, but focused mainly on binary classification tasks
- [Zolfaghari et al., 2025](#) - Proposed a hybrid dual-CNN and ensemble model for MRI-based AD stage classification, improving accuracy but increasing model complexity.
- [Recent 3D CNN Study, 2026](#) - Proposed a multi-modal 3D CNN for Alzheimer's classification using full MRI volumes, improving spatial understanding but with moderate

RESEARCH GAP

- Existing studies primarily focus on improving deep learning architectures or feature extraction techniques, but lack systematic optimization of key model components such as activation functions, optimizers, and loss functions
- Additionally, limited work addresses multi-class classification of EMCI, LMCI, and MCI using MR images

METHODOLOGY

Database

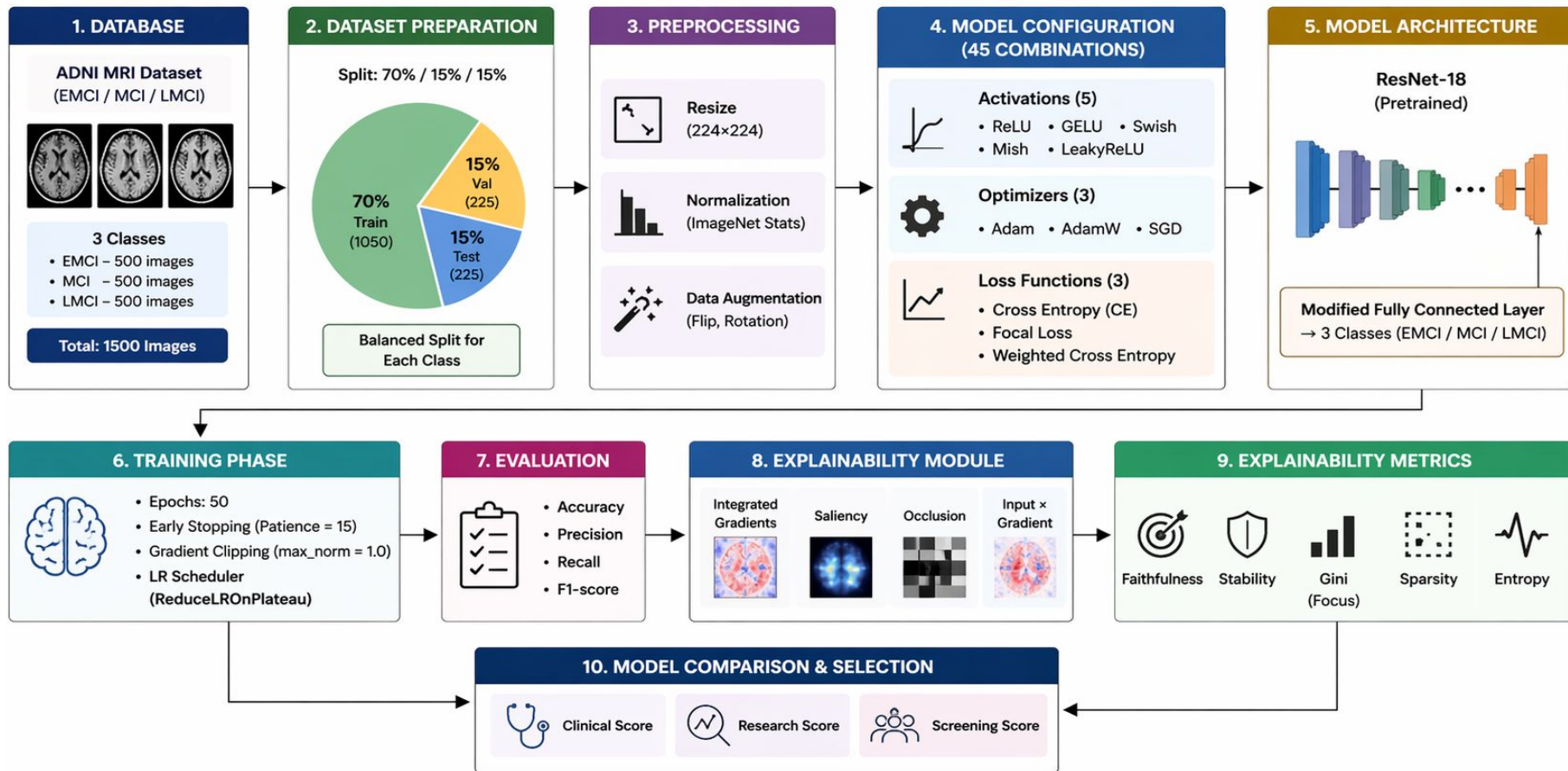
- Dataset: **Alzheimer's Disease Neuroimaging Initiative (ADNI)**
- Type: **Neuroimaging dataset for Alzheimer's research**
- Modality: **T1-weighted MRI scans**
- Total Images : **1500**

Data Split

- Training: **70% (1050 images)**
- Validation: **15% (225 images)**
- Testing: **15% (225 images)**



PROPOSED METHODOLOGY WORKFLOW



The study explore a combinatorial search space using 5 **activation functions**, 3 **optimizers**, and 3 **loss functions**, resulting in a total of 45 model configurations.

Experiment	Optimizer	Activation	Loss Function	Purpose
Adam_ReLU	Adam	ReLU	Cross-Entropy	Baseline model
Adam_Mish	Adam	Mish	Cross-Entropy	Activation comparison
Adam_GELU	Adam	GELU	Cross-Entropy	Activation comparison
Adam_Swish	Adam	Swish	Cross-Entropy	Activation comparison
Adam_ReLU_Focal	Adam	ReLU	Focal Loss	Handle class imbalance
Adam_ReLU_Weighted	Adam	ReLU	Weighted CE	Class balancing
AdamW_ReLU	AdamW	ReLU	Cross-Entropy	Optimizer comparison
SGD_ReLU	SGD	ReLU	Cross-Entropy	Optimizer baseline

- A pretrained **ResNet-18 model** is used for feature extraction
- The model is trained for **50 epochs** using early stopping, gradient clipping, and a learning rate scheduler to ensure stable convergence.

Parameter	Value
Batch Size	32
Initial Learning Rate	0.001
Maximum Epochs	50
Early Stopping Patience	15
Gradient Clipping	max_norm = 1.0
LR Scheduler	ReduceLROnPlateau
Scheduler Details	factor = 0.5, patience = 5
Cross-Validation	10-fold

Explainability Module

- Applied multiple XAI methods to **visualize model decisions on MRI images**
- Generated heatmaps using Integrated Gradients, Saliency Maps, Occlusion, Input $\times$ Gradient
- Evaluated each method highlights in identifying the **important brain regions**
- This helps to understand **the biomarker regions influenced Alzheimer stage classification**

Explainability Metrics

- Computed quantitative metrics for each model:
 - **Faithfulness:** Checked if predictions change when important regions are removed
 - **Stability:** Tested if explanations remain consistent under noise
 - **Gini (Focus):** Measured how concentrated the attention is
 - **Sparsity:** Checked if irrelevant regions are ignored
 - **Entropy:** Measured uncertainty in explanations
- Used these metrics to **compare models and select the most reliable and interpretable one**

RESULTS

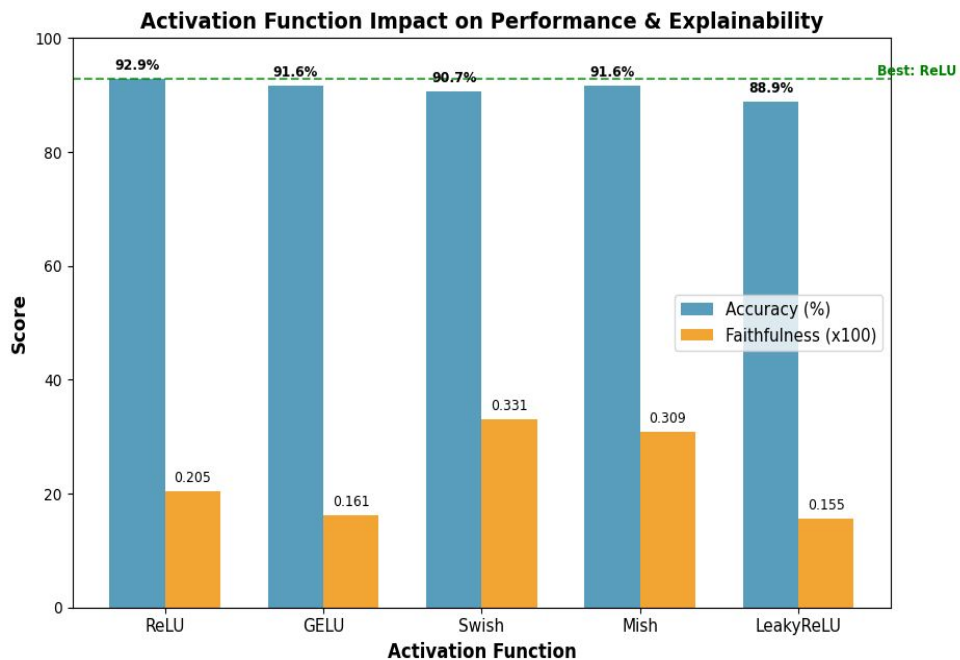
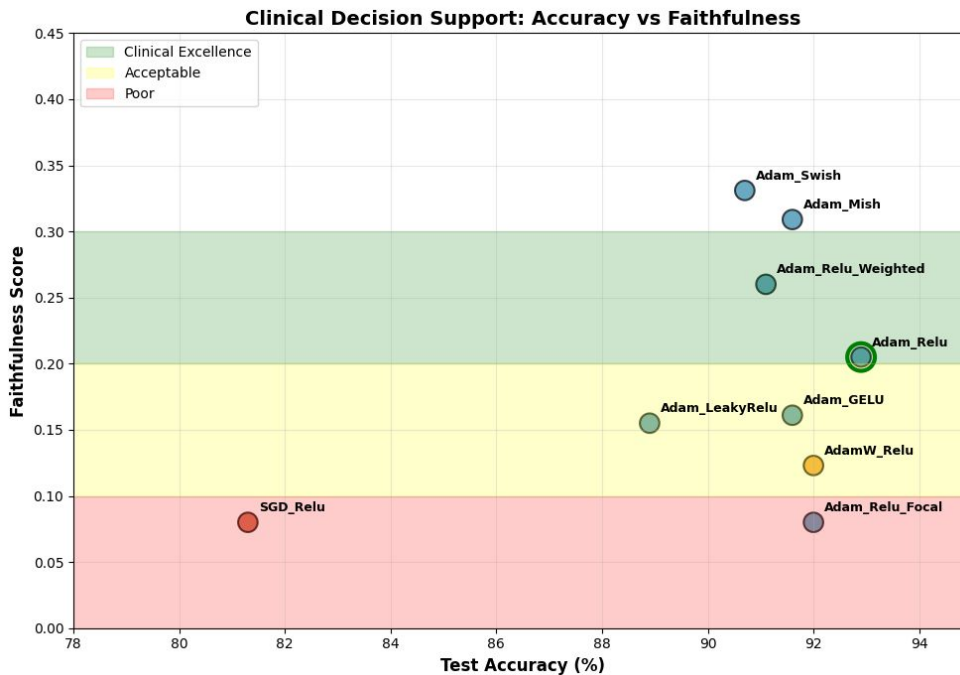
Model	Accuracy	F1 Score	Faithfulness	Stability
Adam_Relu	92.9%	0.929	0.205 (GOOD)	0.189 (GOOD)
AdamW_Relu	92.0%	0.920	0.123 (FAIR)	0.220 (GOOD)
Adam_Relu_Focal	92.0%	0.920	0.080 (POOR)	0.069 (FAIR)
Adam_Mish	91.6%	0.916	0.309 (EXCELLENT)	0.258 (HIGH)
Adam_GELU	91.6%	0.916	0.161 (FAIR)	0.331 (HIGH)
Adam_Relu_Weighted	91.1%	0.911	0.260 (GOOD)	0.206 (GOOD)
Adam_Swish	90.7%	0.906	0.331 (EXCELLENT)	0.347 (HIGH)
Adam_LeakyRelu	88.9%	0.889	0.155 (FAIR)	0.204 (GOOD)
SGD_Relu	81.3%	0.813	0.080 (POOR)	0.076 (FAIR)

BEST MODEL: Adam_Relu

Model	Focus (Gini)	Clinical Score	Research Score	Screening Score
Adam_Relu	0.621 (GOOD)	0.579	0.623	0.850
AdamW_Relu	0.672 (GOOD)	0.433	0.546	0.697
Adam_Relu_Focal	0.668 (GOOD)	0.428	0.482	0.548
Adam_Mish	0.635 (GOOD)	0.677	0.667	0.808
Adam_GELU	0.648 (GOOD)	0.839	0.813	0.894
Adam_Relu_Weighted	0.730 (GOOD)	0.009	0.011	0.001
Adam_Swish	0.606 (GOOD)	0.622	0.740	0.716
Adam_LeakyRelu	0.692 (GOOD)	0.962	0.942	0.865
SGD_Relu	0.431 (FAIR)	0.185	0.277	0.646



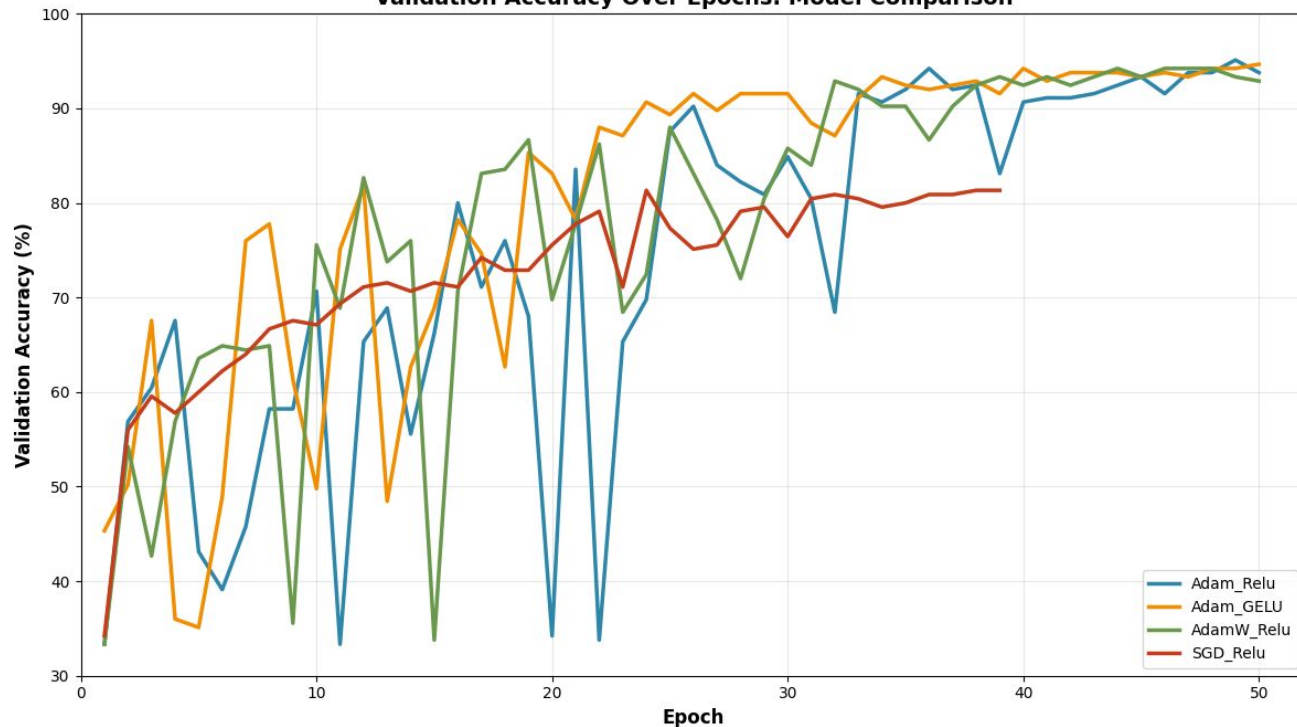
Explainability Analysis





Training Convergence

Validation Accuracy Over Epochs: Model Comparison



The graph shows that Adam_Relu converges fastest to the highest accuracy (95%), making it the optimal choice

CONCLUSION

- Developed an optimized deep learning model for Alzheimer's stage classification using MRI
- Adam + ReLU achieved the best performance with highest accuracy and F1-score
- Explainability analysis ensured reliable and interpretable predictions
- Demonstrated that component optimization alone can significantly improve model performance

ICMLAI-2026

International Conference on
Machine Learning & Artificial Intelligence

Advancing Intelligent Systems. Empowering the Future.

Thank you

We appreciate your time and attention.
Questions & Discussions are welcome!

